Unit: 6 Unit 6: Systems of Linear Equations and Inequalities
Topic: Solving Systems by Graphing

Name: ____________
Date: ________

Advanced (Habanero)

Q1. What is the graph of the system $x+y=2$ and $x-y=-1$?
(A) Two intersecting lines
(B) Two parallel lines
(C) One line
(D) No lines
(E) A circle

Q2. The system $y=2x-1$ and $y=-x+3$ has a solution at
(A) $(0,0)$
(B) $(1,1)$
(C) $(2,3)$
(D) $(-1,-1)$
(E) $(1,2)$

Q3. Which of the following systems has no solution?
(A) $x+y=2$ and $x+y=3$
(B) $x-y=1$ and $x+y=2$
(C) $2x+2y=4$ and $x+y=2$
(D) $y=2x-1$ and $y=2x+1$
(E) $x=2$ and $y=3$

Q4. The solution to the system $y=x+1$ and $y=-x-1$ is
(A) $(0,0)$
(B) $(1,2)$
(C) $(-1,0)$
(D) $(0,-1)$
(E) $(1,0)$

Q5. What is the x-coordinate of the solution to $x+2y=4$ and $3x-2y=5$?
(A) -1
(B) 0
(C) 1
(D) 2
(E) 3

Q6. The system of equations $y=x^2$ and $y=2x-1$ has solutions at
(A) $(0,0)$ and $(1,1)$
(B) $(1,1)$ and $(-1,-1)$
(C) $(1,1)$ and $(2,3)$
(D) $(-1,0)$ and $(1,2)$
(E) $(0,-1)$ and $(2,3)$

Q7. Which of the following is the graph of $x=2$ and $y=-1$?
(A) A point
(B) A line
(C) A circle
(D) A parabola
(E) An ellipse

Q8. What is the y-coordinate of the solution to $y=x+2$ and $y=2x-3$?
(A) -1
(B) 0
(C) 1
(D) 2
(E) 3

Open Ended Questions (FRQ)

Q9. Solve the system $x+y=3$ and $2x-y=1$ graphically. Show your work and find the solution.

Q10. Solve the system $y=x^2-2$ and $y=2x-1$ graphically. Find the x-coordinates of the solutions.

Chef's Tips

- Tip 1: Check for any special cases or exceptions.
- Tip 2: Verify the reasonableness of the solutions.
- Tip 3: Use a graphing tool or software to visualize the systems.